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(54) **FOOD PROCESSOR WITH DICING GRID ACCESSORY AND CLEANING TOOL THEREFOR**

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(52) **U.S. Cl.**

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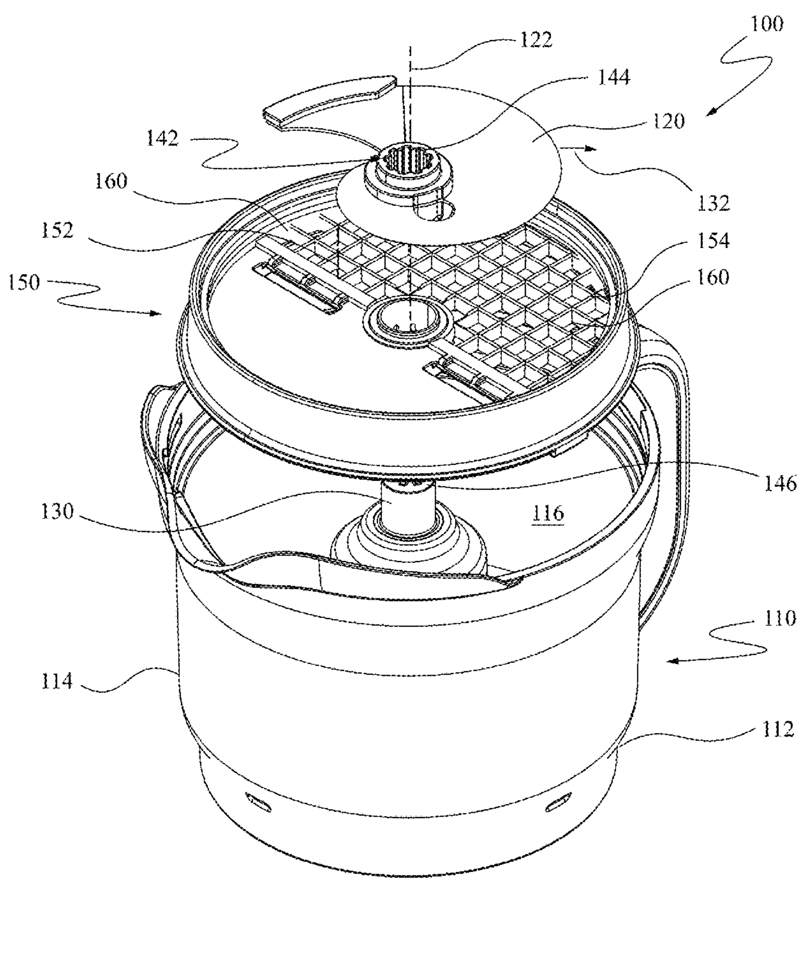
(57) **ABSTRACT**

A cleaning tool for cleaning a dicing grid accessory of a food processor, the dicing grid accessory having a dicing grid with a plurality of apertures, and an anchor hole located adjacent to the dicing grid, the cleaning tool having:

a body having a plurality of bosses dimensioned to be receivable by the plurality of apertures;

an anchor arm extending from the body, the anchor arm having a shape that conforms to the anchor hole of the dicing grid accessory and being releasably engageable with the anchor hole, such that, when the anchor arm is engaged with the anchor hole, the body is located with respect to the dicing grid and is pivotable about the anchor hole to allow the plurality of bosses to move into the plurality of apertures; and

a handle extending from the body opposite the anchor arm, such that, when the anchor arm is engaged with the anchor hole the handle provides leverage to increase the available force at the plurality of bosses as the bosses travel into the plurality of apertures.



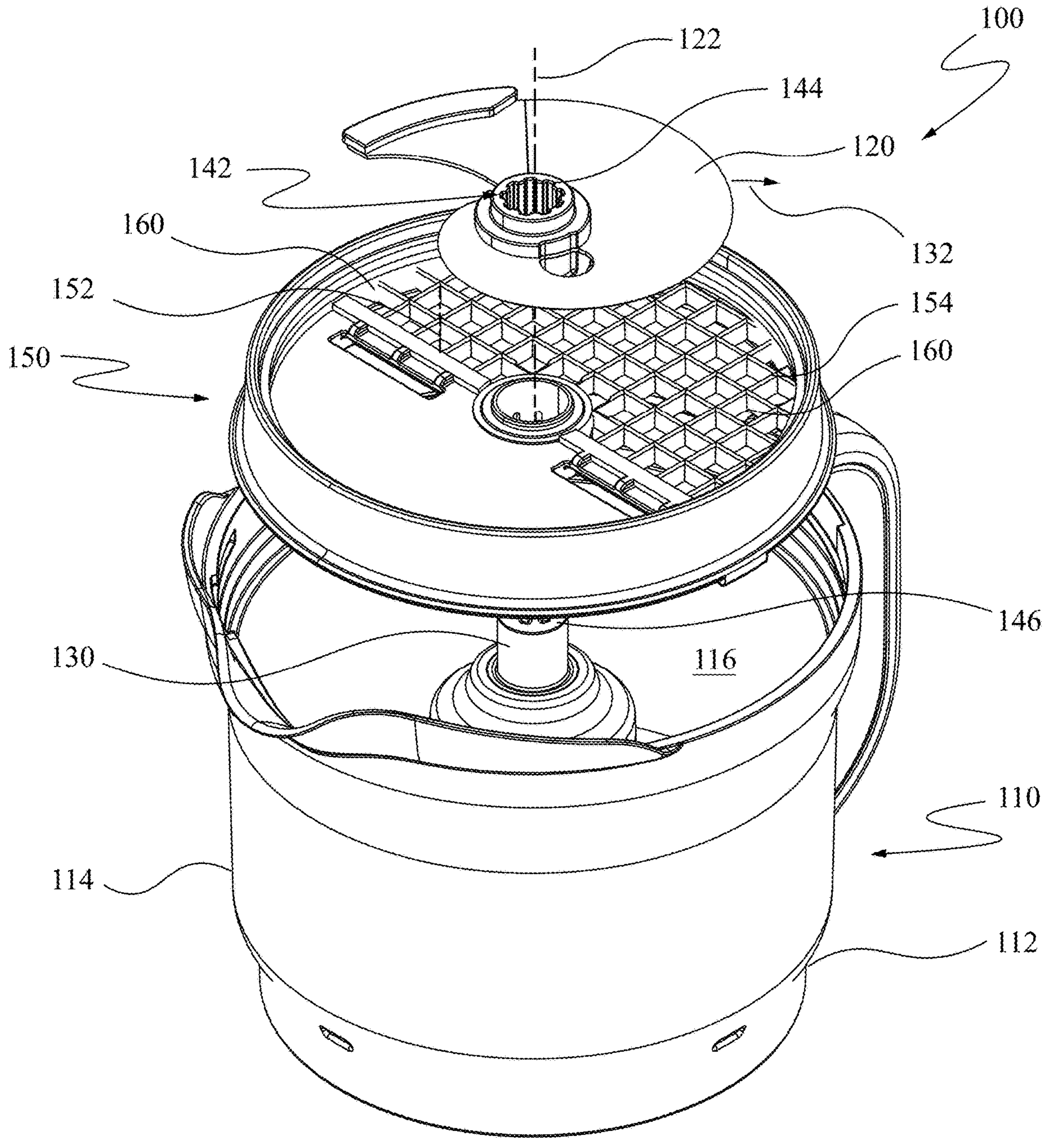

100
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FIG. 1

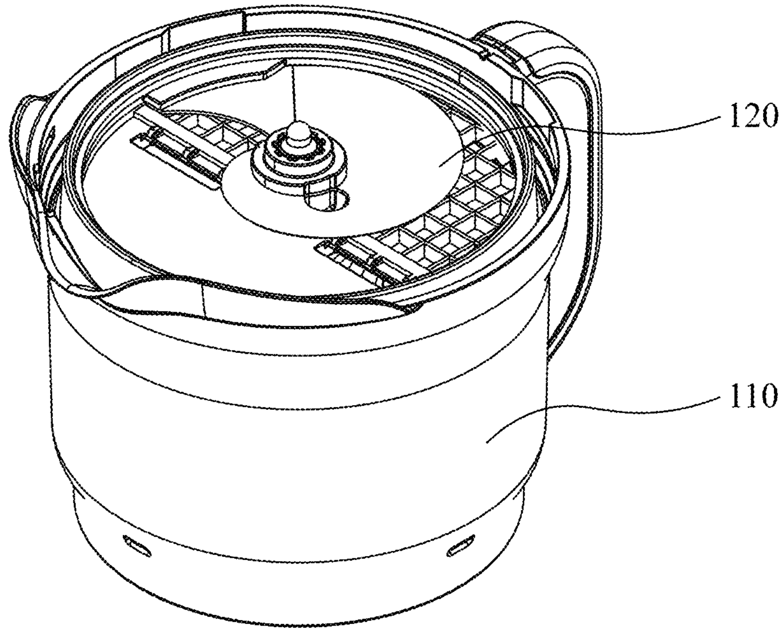

FIG. 2

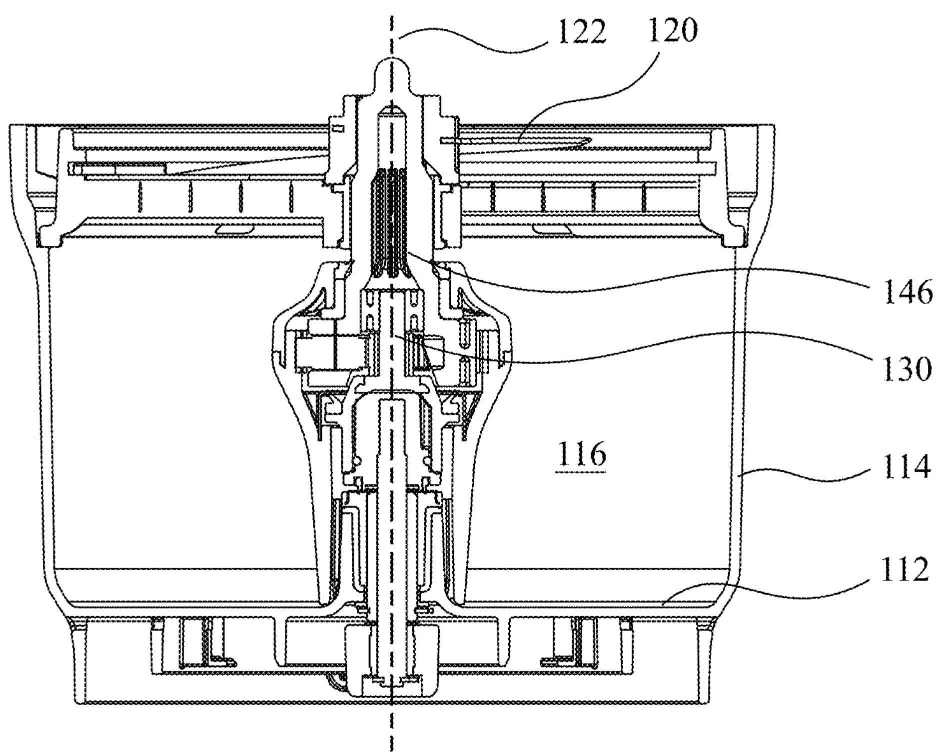

FIG. 3

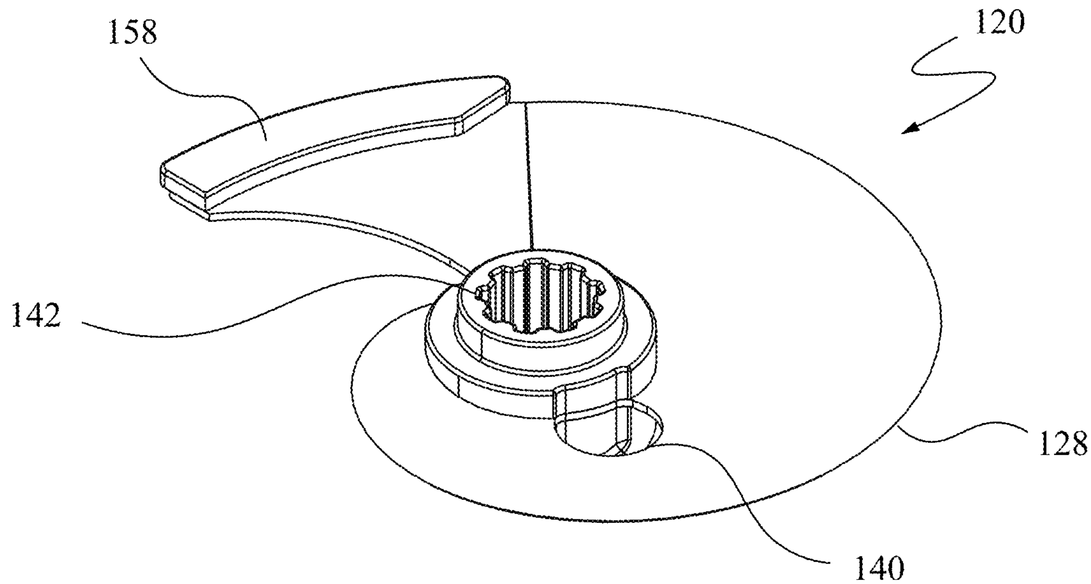

FIG. 4

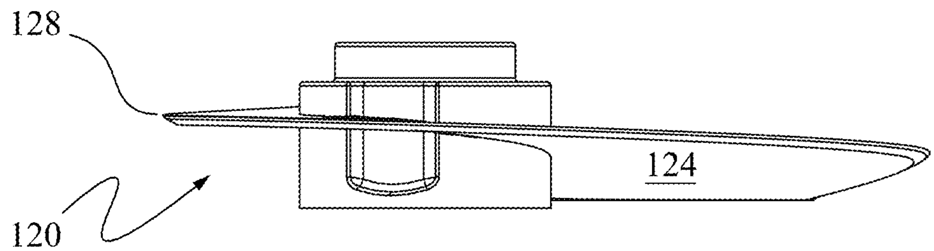

FIG. 5

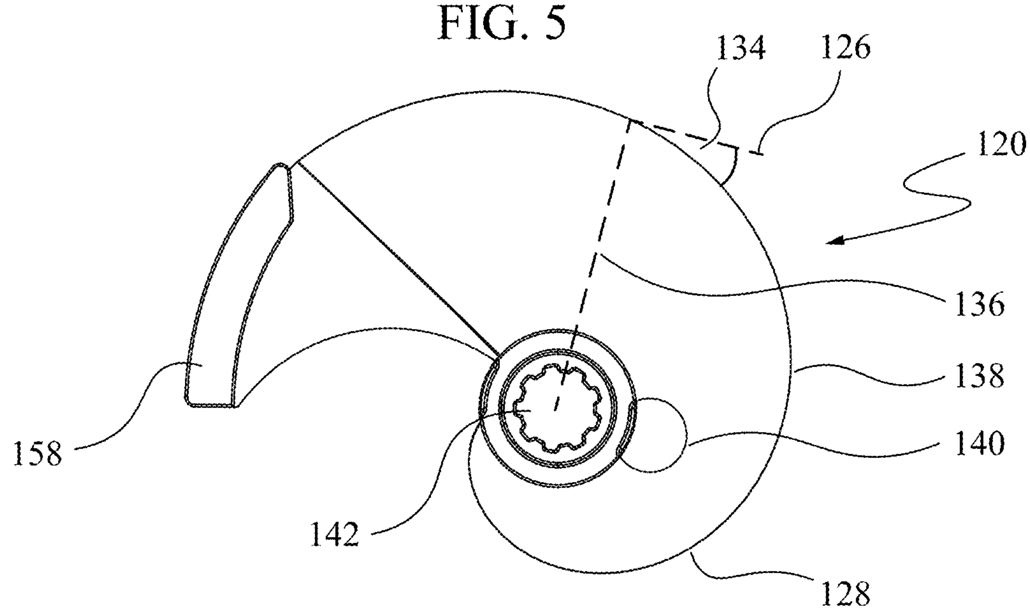

FIG. 6

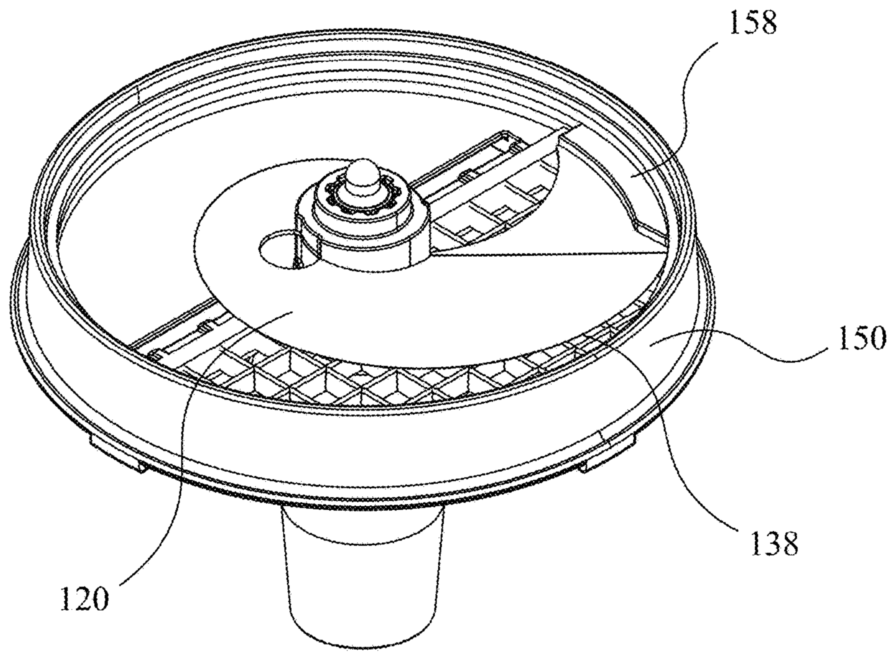

FIG. 7

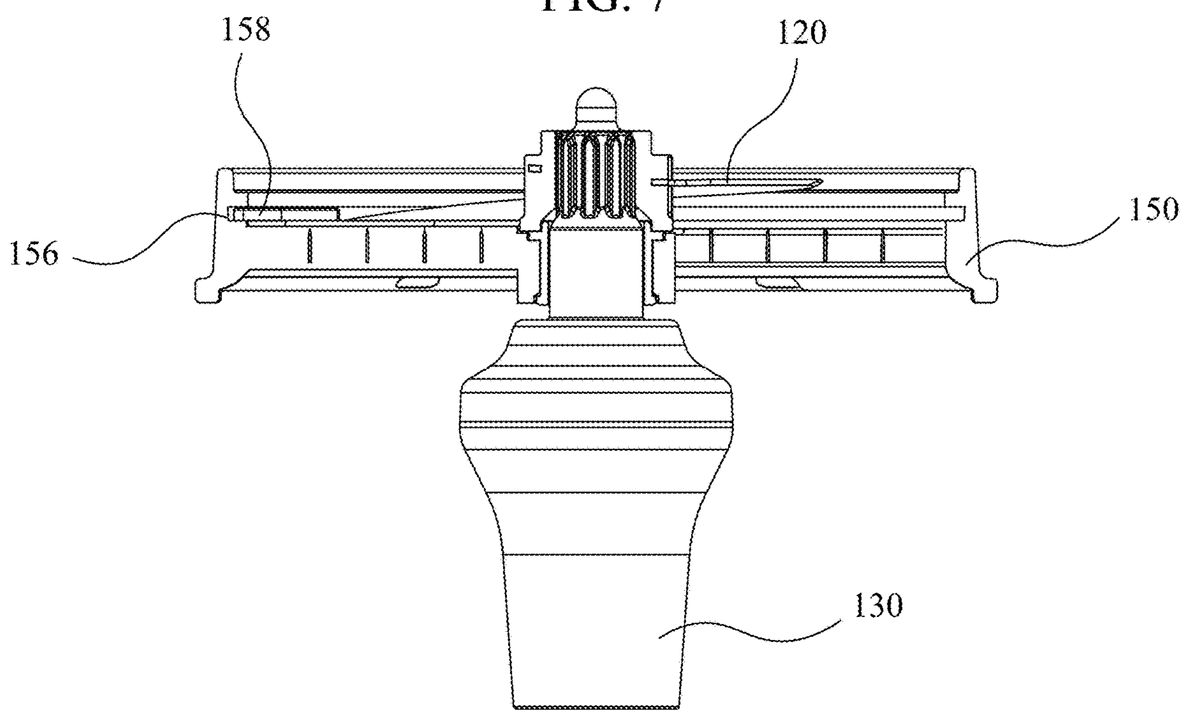

FIG. 8

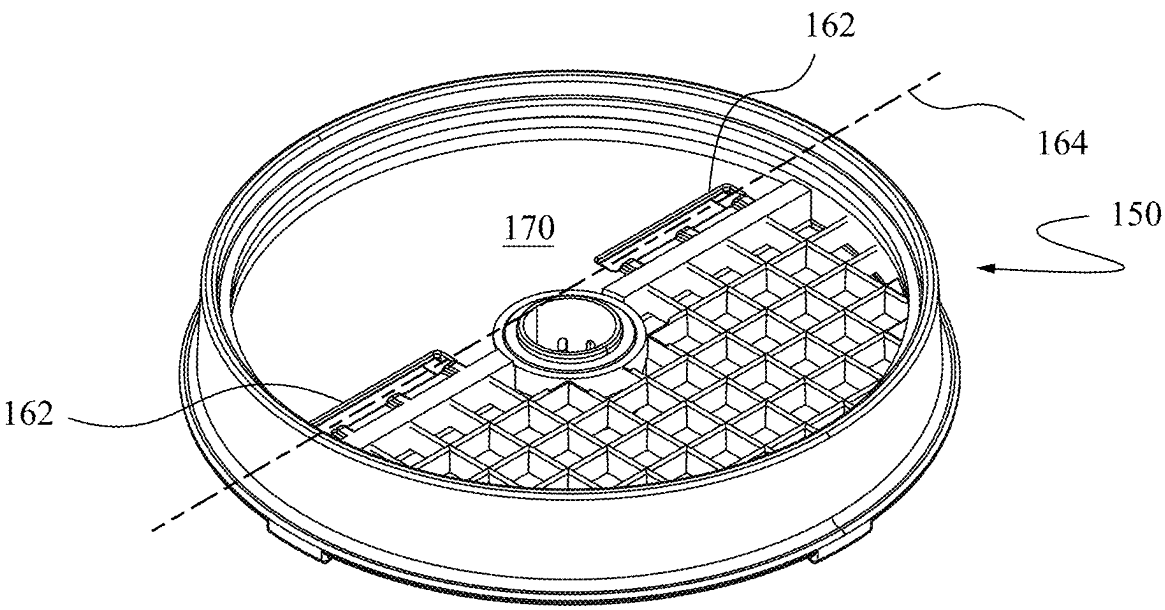
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FIG. 9

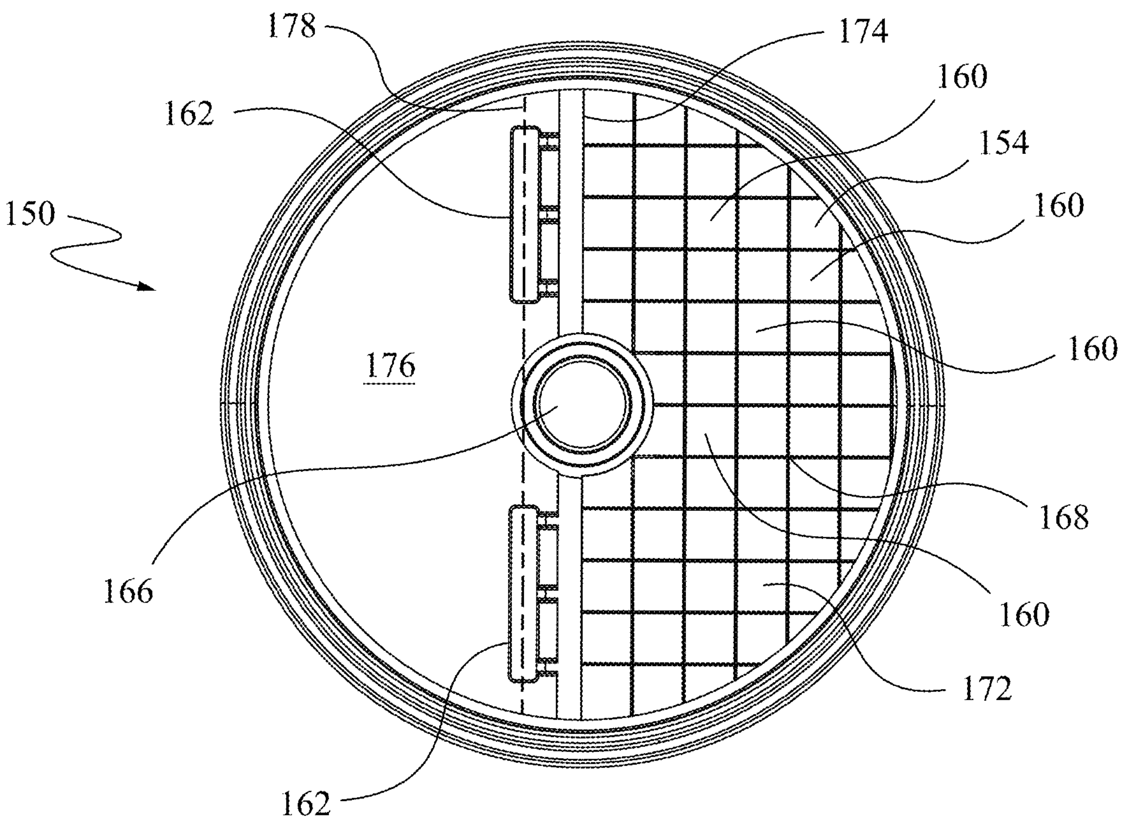
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FIG. 10

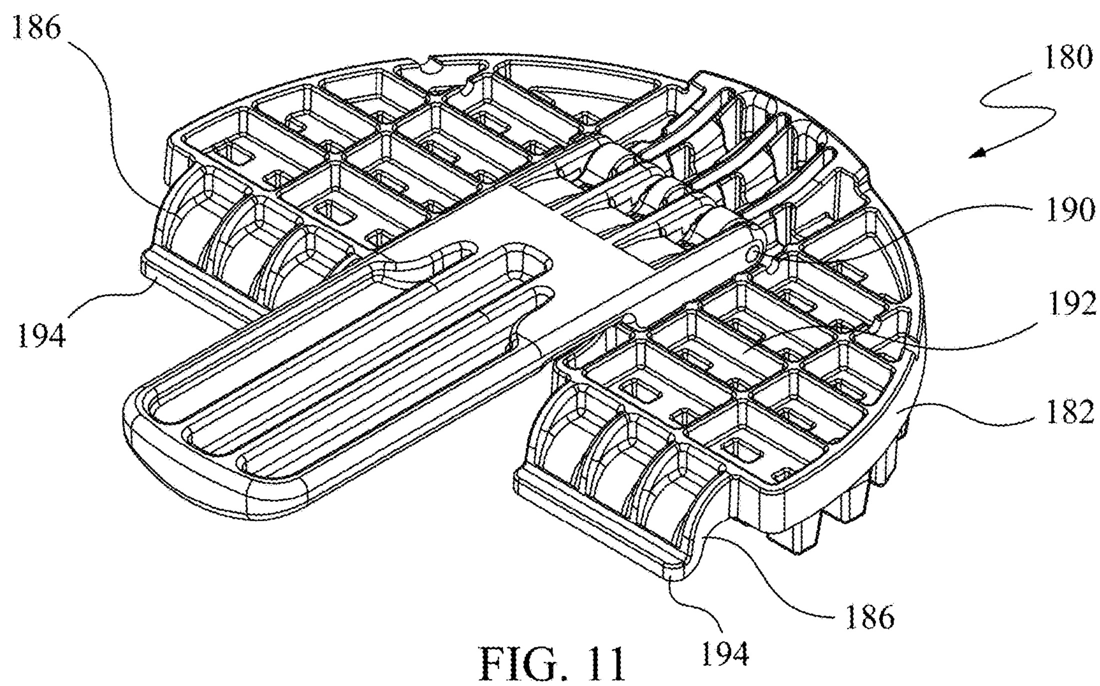

FIG. 11

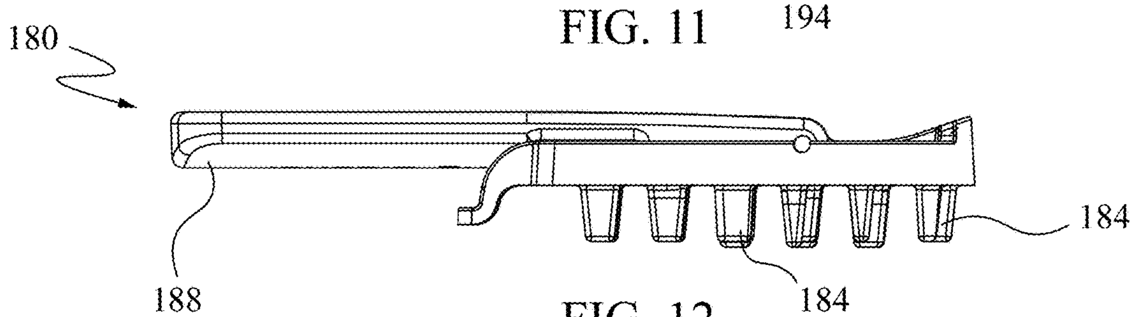

FIG. 12

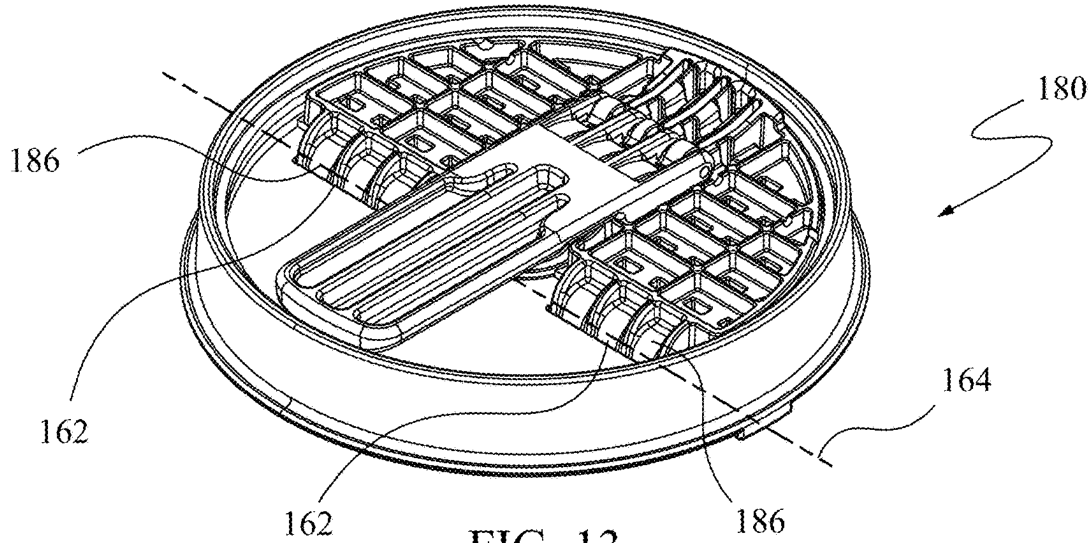

FIG. 13

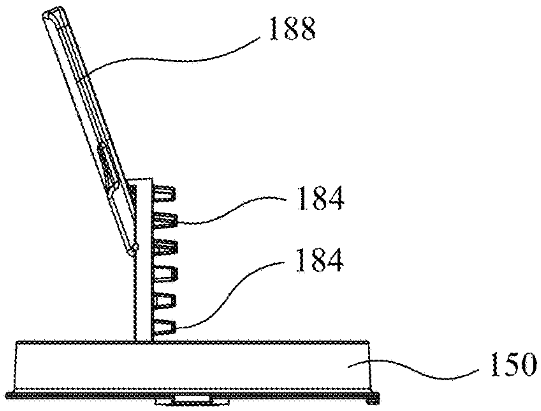

FIG. 14A

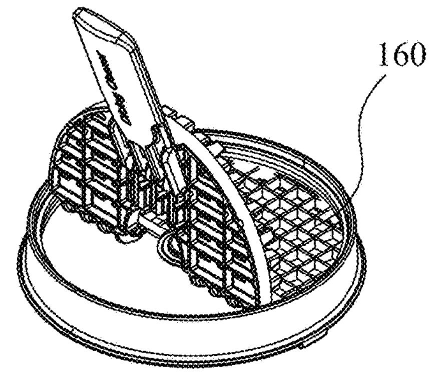

FIG. 14B

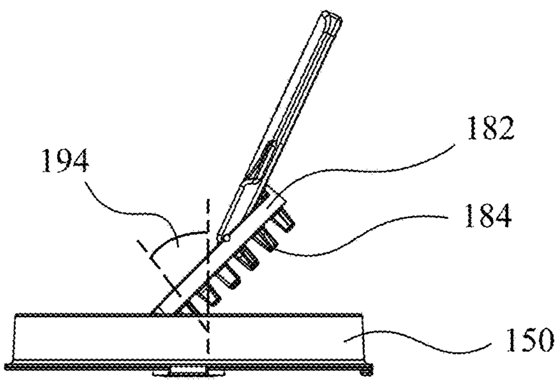

FIG. 15A

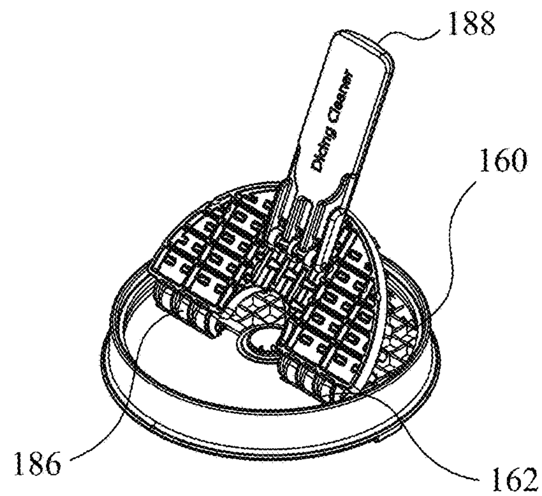

FIG. 15B

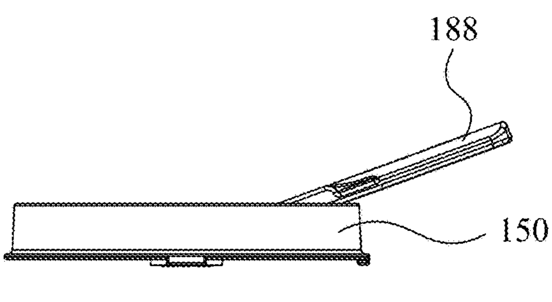

FIG. 16A

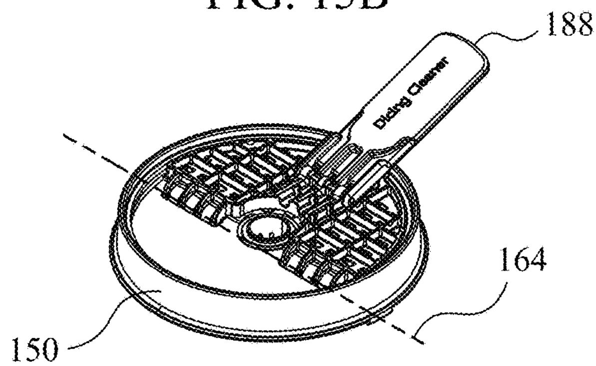

FIG. 16B

FOOD PROCESSOR WITH DICING GRID ACCESSORY AND CLEANING TOOL THEREFOR

RELATED APPLICATIONS

[0001] This application claims convention priority from Australian Provisional Patent Application No. 2022901110, the contents of which are incorporated herein in their entirety by reference thereto.

FIELD

[0002] This invention relates to food processor technology.

BACKGROUND

[0003] Dicing grid accessories are becoming increasingly popular in the use of food processors, in order to process food items into diced food items quickly and efficiently. One issue plaguing dicing grids is that food items, particularly tough, starchy food items such as sweet potato, can become wedged, lodged, or otherwise stuck in the dicing grid and prevent use of the dicing grid. The dicing grid becomes inoperable either because the following food item being processed is softer and thus not able to dislodge the previous food item from the dicing grid, or because the dicing grid is blocked to such an extent that the force required to dislodge the food items from the dicing grid exceeds the permissible torque of the drive train.

[0004] Even smaller blockages are undesirable, as they lead to food items not being sliced cleanly, due to the backpressure imparted by the lodged food items in the dicing grid, and the slow pressing of food items through the dicing grid that offer resistance, rather than smooth slicing at higher velocities without the resistance.

SUMMARY

[0005] It is an object of the present invention to at least substantially address one or more of the above disadvantages, or at least provide a useful alternative to the above approaches.

[0006] In a first aspect there is disclosed a cleaning tool for cleaning a dicing grid accessory of a food processor, the dicing grid accessory having a dicing grid with a plurality of apertures, and an anchor hole located adjacent to the dicing grid, the cleaning tool having:

- **[0007]** a body having a plurality of bosses dimensioned to be receivable by the plurality of apertures;
- **[0008]** an anchor arm extending from the body, the anchor arm having a shape that conforms to the anchor hole of the dicing grid accessory and being releasably engageable with the anchor hole, such that, when the anchor arm is engaged with the anchor hole, the body is located with respect to the dicing grid and is pivotable about the anchor hole to allow the plurality of bosses to move into the plurality of apertures; and
- **[0009]** a handle extending from the body opposite the anchor arm, such that, when the anchor arm is engaged with the anchor hole the handle provides leverage to increase the available force at the plurality of bosses as the bosses travel into the plurality of apertures.

[0010] Preferably, the body includes a hinge on a surface opposite the plurality of bosses, and the handle is attached to the body through the hinge such that the handle is moveable between a rest position, wherein the handle extends from the hinge in a direction toward the anchor arm so as to reduce a total length of the cleaning tool, and an operational position, wherein the handle extends from the hinge in a direction opposite the anchor arm, and a storage position.

[0011] Preferably, each boss in the plurality of bosses is chamfered to allow the boss to engage the respective aperture at an angle while the cleaning tool is being pivoted to move the boss into the aperture.

[0012] Preferably, the anchor arm includes a hook to allow the anchor arm to engage the anchor hole.

[0013] In a second aspect there is disclosed a dicing grid accessory for a food processor to be cleaned by the cleaning tool of the first aspect, the dicing grid accessory having:

- **[0014]** a dicing grid with a plurality of apertures; and
- **[0015]** an anchor hole located adjacent to the dicing grid.

[0016] Preferably, the anchor hole includes a linear slot.

[0017] Preferably, the anchor hole includes two or more linear slots located along a common anchor direction.

[0018] Preferably, the dicing grid accessory includes an opening for receiving a drive shaft of the food processor, wherein the dicing grid is on a first side of the dicing grid accessory, and wherein the linear slot is on a second side of the dicing grid accessory along an anchor direction.

[0019] Preferably, the dicing grid accessory has a circular shape, with the opening in the center, and the anchor direction is a non-diametric chord of the dicing grid accessory.

[0020] In a third aspect, there is disclosed a food processor to process a food item into a diced food item, the food processor having:

- **[0021]** a vessel having a floor, sidewalls extending upwardly from the floor to define a space for receiving the diced food item;
- **[0022]** a blade mounted to the vessel so as to be rotatable about a drive axis the food item;
- **[0023]** a drive shaft for driving the blade about the drive axis to slice the food item in a first direction; and
- **[0024]** a dicing grid mounted to the vessel below the blade for sectioning the sliced food item to create the diced food item,
- **[0025]** wherein the dicing grid includes a circular track and the blade includes at a perimeter thereof a cam that is engageable with the track such that a force path exists from the blade to the dicing grid.

[0026] Preferably, a bottom surface of the blade is non-parallel to the dicing grid to create a gap between a leading edge of the blade and the dicing grip, such that the sliced food item is urged into the dicing grid by the bottom surface of the blade as the blade is rotated by the drive shaft.

[0027] Preferably, the bottom surface of the blade has a helical shape.

[0028] Preferably, the blade has a leading edge that first engages the food item for slicing, wherein the leading edge has a negative angle of attack relative to a radius of the blade to reduce the force imparted from the food item on the blade as the blade slices the food item.

[0029] Preferably, the leading edge is continuously curved relative to the radius of the blade in a spiral shape such that the perimeter of the blade forms a single continuous leading edge.

[0030] Preferably, the spiral shape is a Fibonacci spiral.

[0031] Preferably, the blade includes a lift aperture to allow access for a finger of a user to lift the blade from the vessel.

[0032] Preferably, the lift aperture is located adjacent the drive shaft.

[0033] There is also disclosed, in combination, the cleaning tool of the first aspect and the dicing grid accessory of the second aspect.

BRIEF DESCRIPTION OF THE DRAWING

[0034] Preferred embodiments of the present invention will now be described by way of example, with reference to the accompanying drawings, wherein:

[0035] FIG. **1** is an exploded perspective view of a food processor according to a preferred embodiment of the invention.

[0036] FIG. **2** is a perspective view of the food processor of FIG. **1**.

[0037] FIG. **3** is a vertically sectioned side elevation view of the food processor of FIG. **1**.

[0038] FIG. **4** is a detailed perspective view of a blade used with the food processor of FIG. **1**.

[0039] FIG. **5** is a side elevation view of the blade of FIG. **4**.

[0040] FIG. **6** is a top plan view of the blade of FIG. **4**.

[0041] FIG. **7** is a detailed perspective view of the food processor of FIG. **1**.

[0042] FIG. **8** is a vertically sectioned side elevation view of the food processor of FIG. **7**.

[0043] FIG. **9** is a detailed perspective view of a dicing grid accessory used with the food processor of FIG. **1**.

[0044] FIG. **10** is a top plan view of the dicing grid accessory of FIG. **9**.

[0045] FIG. **11** is a detailed perspective view of a cleaning tool for the dicing grid accessory of FIG. **9**.

[0046] FIG. **12** is a side elevation view of the cleaning tool of FIG. **11**.

[0047] FIG. **13** is a perspective view of the cleaning tool of FIG. **11** used with the dicing grid accessory of FIG. **9** with a handle in a rest position.

[0048] FIG. **14**A is a side elevation view of the cleaning tool and dicing grid accessory of FIG. **13** being pivoted to clean the dicing grid accessory.

[0049] FIG. **14**B is a perspective view of the cleaning tool and dicing grid accessory of FIG. **13** being pivoted to clean the dicing grid accessory.

[0050] FIG. **15**A is a side elevation view of the cleaning tool and dicing grid accessory of FIG. **13** being pivoted to clean the dicing grid accessory.

[0051] FIG. **15**B is a perspective view of the cleaning tool and dicing grid accessory of FIG. **13** being pivoted to clean the dicing grid accessory.

[0052] FIG. **16**A is a side elevation view of the cleaning tool and dicing grid accessory of FIG. **13** being pivoted to clean the dicing grid accessory.

[0053] FIG. **16**B is a perspective view of the cleaning tool and dicing grid accessory of FIG. **13** having completed the cleaning operation.

DETAILED DESCRIPTION

[0054] FIG. **1** shows a food processor **100** according to a preferred embodiment of the invention. Not shown in this embodiment are the drive unit consisting of motor, gearbox, and user interface, or the lid, chute, or food pusher, since these elements of food processors are well-known and do not form part of the inventive concepts disclosed herein.

[0055] The food processor **100** for processing a food item (not shown) into a diced food item (not shown) includes a vessel **110** having a floor **112** and one or more sidewalls **114** extending upwardly from the floor to define a space **116**. The space **116** is typically sized to accommodate the diced food item. The food processor **100** further includes a blade **120** mounted to the vessel **110**, as shown in FIG. **3**, so as to be rotatable about a drive axis **122** to slice the food item. The food processor **100** further includes a drive shaft **130** for driving the blade **120** about the drive axis **122** in a first direction **132** to slice the food item. The blade **120** is connected to the drive shaft **130** by an opening **142** having a cammed profile **144** that matches a cammed profile **146** of the drive shaft **130**.

[0056] As shown in FIGS. **4** to **6**, the blade **120** further has a leading edge **128**, which is the part of the blade **120** that first comes into contact with the food item as the food item is being sliced by the blade **120** in the first direction **132** to create a sliced food item (not shown). The leading edge **128** preferably has a negative angle of attack **134** relative to an azimuthal direction **126**, or tangent to the perimeter **138**, of the blade **120** as the leading edge **128** engages the food item. Preferably, as best seen in FIG. **6**, the leading edge **128** is continuously curved relative to a radius **136** of the blade **120** in a spiral shape, such that a perimeter **138** of the blade **120** forms a single continuous leading edge **128**. Preferably, the spiral shape is a Fibonacci, or in a preferred embodiment a logarithmic, spiral.

[0057] The blade **120** further has a bottom surface **124** that slopes downwardly in the azimuthal direction **126**. The downwardly sloping bottom surface **124** urges the food item, after it was sliced by the leading edge **128**, downwardly due to the blade **120** being fixed to the drive shaft **130** and impingement of the food item against the bottom surface **124**. Preferably, the bottom surface **124** has a helical shape as best seen in FIG. **5**. Preferably, the blade **120** is of constant thickness and thin, such that the sloping of the bottom surface **124** is equivalent to the shape of the blade **120**.

[0058] As best shown in FIG. **6**, the blade **120** further includes a lift aperture **140** to allow access for a finger of a user (not shown) to lift the blade **120** from the drive shaft **130** and/or vessel **110**. Preferably, as shown in FIG. **6**, the lift aperture **140** is located adjacent to the opening **142** and preferably has a substantially circular shape.

[0059] Returning briefly to FIG. **1**, the food processor **100** further includes a dicing grid accessory **150** that is mounted to the vessel **110** below the blade **120**. The dicing grid accessory **150** includes a dicing grid **154** having a plurality of apertures **160** for sectioning the sliced food item along a second direction **152** to create the diced food item. For example, the upper edges of the apertures **160** may be thin and/or sharpened. As shown in FIGS. **7** and **8**, the dicing grid accessory **150** includes a circular track **156** that may be formed as a groove or simply as an inward protrusion. The blade **120** includes a cam **158** at the perimeter **138** of the blade **120**. The cam **158** is engageable with the track **156** such that a force path exists from the blade **120** to the dicing grid accessory **150**. Preferably, the cam **158** has a thickness that is larger than a thickness of the blade **120**.

[0060] Moving now to FIGS. **9** and **10** to discuss the dicing grid accessory **150** in more detail. The dicing grid accessory includes an anchor hole **162** located adjacent to the dicing grid **154**. Preferably, as shown in FIG. **10**, the anchor hole includes a linear slot **162**, more preferably two or more linear slots **162** located along a common anchor direction **164**. The dicing grid accessory **150** further includes an opening **166** dimensioned to be larger than the cammed profile **144** of the drive shaft **130** so that the drive shaft **130** is allowed to rotate freely within the opening **166**. Preferably, the dicing grid **154** only covers a portion **168** of a circular surface **170** of the preferably circular shaped dicing grid accessory **150**, being a first side **172**. In the embodiment shown in FIGS. **9** and **10**, the portion **168** is defined by a first chord **174** that preferably intersects at least a portion of the opening **166**. The linear slot **162** is located on a second side **176**, being on an opposite side of the surface **170** than the first side **174**. Preferably, the common anchor direction **164** is parallel to the first chord **174** and is located along a second chord **178** that is non-diametric, i.e., it does not intersect the center of the surface **170**, and more preferably does not intersect the opening **166**.

[0061] Moving now to FIGS. **11** to **13**, which show a cleaning tool **180** for cleaning the dicing grid accessory **150** of the food processor **100**. The cleaning tool includes a flat body **182** having a plurality of bosses **184** or teeth dimensioned to be receivable by the plurality of aperture **160**.

[0062] The cleaning tool **180** further includes an anchor arm **186** extending from the flat body **182** and having a shape that conforms to the anchor hole **162** of the dicing grid accessory **150**.

[0063] The anchor arm **186** is realeasably engageable with the anchor hole **162**, for example by inclusion of a hook **194** to allow the anchor arm **186** to engage the anchor hole **162** and then bear against the surface **170** of the dicing grid accessory **150** to resist torque applied to the cleaning tool **180**. The anchor arm **186** and anchor hole **162** are further situated such that, when the arm **186** is engaged with the hole **162**, the flat body **182** is located with respect to the dicing grid **154** and is pivotable about the anchor hole **162** to allow the bosses **184** to move into the apertures **160**. Thus, when the arm **186** is engaged with the hole **162**, the cleaning tool **180** has a single degree of freedom with respect to the dicing grid accessory **150**, the single degree of freedom being hinging about the anchor hole **162** or common anchor direction **164**.

[0064] The cleaning tool **180** further includes a handle **188** extending from the flat body **182** opposite the anchor arm **186**, as shown in FIG. **16**A. Thus, when the anchor arm **186** is engaged with the anchor hole **162**, the handle **188** provides leverage in the hinging motion about the common anchor direction **164** to increase the available force at the bosses **184** as the bosses **184** travel into the plurality of apertures **160**. Returning to FIG. **11**, the flat body **182** preferably includes a hinge **190** on a top surface **192** opposite the bosses **184**. The handle **188** is attached to the flat body **182** using the hinge **190** such that the handle **188** is movable about the hinge **190** between a rest position, shown in FIGS. **11** to **13**, and an operational position, shown in FIGS. **14**A to **16**B. In an alternative embodiment, the handle **188** can be removably attached via a ball and socket joint (not shown). In the rest position, the handle **188** extends from the hinge in a direction toward the anchor arm **186**, while in the operational position the handle **188** extends from the hinge in a direction away from the anchor arm **186**.

[0065] As shown in FIG. **12**, each boss **184** in the plurality of bosses **184** is preferably chamfered, or angled, or rounded, to allow the boss **184** to engage the respective aperture **160** at an angle **194**, shown in FIG. **15**A, while the cleaning tool **180** is being pivoted to move the boss **184** into the aperture **160**.

[0066] Use of the food processor **100**, blade **120**, and cleaning tool **180** will now be discussed.

[0067] To use the food processor **100** with the dicing grid accessory **150**, the dicing grid accessory **150** is mounted on the vessel **110** such that the drive shaft **130** protrudes through the opening **166** in the dicing grid. The blade **120** is then mounted on the drive shaft **130** such that the cammed profiles **144**, **146** engage to allow the transmission of torque from the drive shaft **130** to the blade **120**. The cam **158** of the blade **120** is inserted into the track **156** by flexing the blade **120** temporarily, allowing the cam **158** to be inserted into the track **156**. The food item may now be processed into a diced food item.

[0068] The food item is introduced to the food processor **100**, typically using a chute or similar feed mechanism (not shown). The food item is engaged by the leading edge **128** of the blade **120**. If the food item is tough to slice, the angle of attack **134** of the leading edge **128** allows the blade **120** to keep rotating while progressively slicing the food item. Once the food item has been sliced, the blade **120** rotates over the food item and keeps rotating. The downwardly sloping bottom surface **124** of the blade thus urges the food item into the dicing grid **154**. Preferably, the food item is at least partially engaged by the dicing grid **154**, such that the food item does not move substantially in response to the rotation of the blade **120**, but is restricted to moving downwards into the dicing grid **154** under the urging of the blade **120**. The dicing grid **154** sections the sliced food item to produce the diced food item, which subsequently collects in the vessel **110**.

[0069] Should the dicing grid **154** become blocked, or a food item having a different hardness or consistency to a previous food item is to be processed, the dicing grid **154** may be cleaned using the cleaning tool **180**. To do so, the blade **120** is removed from the drive shaft **130**, and the cleaning tool **180** is engaged to the dicing grid by inserting the anchor arms **186** into the anchor holes **162**, using the hooks **194**. This restricts movement of the cleaning tool **180** relative to the dicing grid **154** to hinging of the cleaning tool about the common anchor direction **164**. Preferably, the handle **188** of the cleaning tool **180** is moved to the operational position as seen in FIG. **14**A. The insertion of the anchor arms **186** into the anchor holes **162** is also easier if the handle is in the operational position before engaging the arms **186** into the holes **162**.

[0070] The cleaning tool **180** is then hinged about the common anchor direction **164** as seen in FIGS. **14**A to **16**B. The progressive engagement of apertures **160** by bosses **184** in a row-by-row fashion enabled by the hinging motion about the common anchor direction **164** maximises the force applied to the blockage in each aperture **160** from the torque applied to the handle **188**, compared to, for example, a cleaning tool that approaches the apertures **160** linearly from above.

[0071] For storage of the food processor **100**, the handle **188** is returned to the rest position and may be stored with the dicing grid accessory **150**, as shown in FIG. **13**.

[0072] Advantages of the food processor **100**, blade **120**, and cleaning tool **180** will now be discussed.

[0073] The motion of the handle **188** between the rest position and the operational position allows the cleaning tool **180** to be stored with the dicing grid accessory **150**, while increasing the available torque for cleaning of the apertures **160** when in the operational position.

[0074] The chamfering or rounding of the bosses **184** allows entry of the bosses **184** into the apertures **160** at an angle, decreasing the chances of the cleaning tool **180** jamming during use, and reducing the manufacturing tolerance requirements for production of the cleaning tool **180**. The use of the anchor arms **186** and anchor holes **162** allows location of the cleaning tool **180** with respect to the dicing grid **154** to ensure that the bosses **184** reliably clean the apertures **160**. The use of two linear slots **162** reduces the degrees of freedom available to the cleaning tool **180** and further increases reliability of its use with the dicing grid **154**. The location of the anchor hole **162** on a side opposite the dicing grid **154** increases the amount of force produced on blockages in the apertures **160** for a given torque.

[0075] The use of the track **156** and cam **158** reduces flexing of the blade **120** and improves the blade's **120** ability to reliably slice the food item. The sloping of the bottom surface **124** improves throughput of the food processor **100** by urging the sliced food item through the dicing grid **154**. The use of the negative angle of attack **134** along the leading edge **128** reduces the torque applied on the drive shaft **130** by the food item as it is sliced by the blade **120**. The use of a single spiral shape, preferably a Fibonacci spiral, maximises the time the blade **120** has at its disposal for successfully slicing a food item until the food item is wedged between the blade **120** and the sidewalls **114** of the vessel **110**, at which point the torque applied by the food item on the drive shaft **130** dramatically increases. The use of the Fibonacci spiral shape ensures that the angle of attack **134** between the blade **120** and the food item is constant as the food item is displaced along a radial **136** direction during the slicing action. The advantage of the blade **120** is that the food item is continuously sliced, thereby distributing and minimizing forces applied to the blade and the food item, as opposed to a straight blade (not shown). The use of the lift aperture **140** allows the blade **120** to be safely removed from the drive shaft **130**. The location of the lift aperture **140** adjacent the opening **142** allows the drive shaft **130** to be used as a bracing point while lifting the blade **120**.

1.-**18**. (canceled)

19. A food processor to process a food item into a diced food item, the food processor having:

- a vessel having a floor, sidewalls extending upwardly from the floor to define a space for receiving the diced food item; a blade mounted to the vessel so as to be rotatable about a drive axis the food item;
- a drive shaft for driving the blade about the drive axis to slice the food item in a first direction; and
- a dicing grid mounted to the vessel below the blade for sectioning the sliced food item to create the diced food item,
- wherein the dicing grid includes a circular track and the blade includes at a perimeter thereof a cam that is engageable with the track such that a force path exists from the blade to the dicing grid.

20. The food processor of claim **19**, wherein a bottom surface of the blade is non-parallel to the dicing grid to create a gap between a leading edge of the blade and the dicing grip, such that the sliced food item is urged into the dicing grid by the bottom surface of the blade as the blade is rotated by the drive shaft.

21. The food processor of claim **20**, wherein the bottom surface of the blade has a helical shape.

22. The food processor of claim **19**, wherein the blade has a leading edge that first engages the food item for slicing, wherein the leading edge has a negative angle of attack relative to a radius of the blade to reduce the force imparted from the food item on the blade as the blade slices the food item.

23. The food processor of claim **22**, wherein the leading edge is continuously curved relative to the radius of the blade in a spiral shape such that the perimeter of the blade forms a single continuous leading edge.

24. The food processor of claim **23**, wherein the spiral shape is a Fibonacci spiral.

25. The food processor of claim **19**, wherein the blade includes a lift aperture to allow access for a finger of a user to lift the blade from the vessel.

26. The food processor of claim **25**, wherein the lift aperture is located adjacent the drive shaft.

27. In combination, the food processor of claim **19**, and a cleaning tool for cleaning a dicing grid accessory of the food processor, the dicing grid accessory having a dicing grid with a plurality of apertures, and an anchor hole located adjacent to the dicing grid, the cleaning tool having:

- a body having a plurality of bosses dimensioned to be receivable by the plurality of apertures;
- an anchor arm extending from the body, the anchor arm having a shape that conforms to the anchor hole of the dicing grid accessory and being releasably engageable with the anchor hole, such that, when the anchor arm is engaged with the anchor hole, the body is located with respect to the dicing grid and is pivotable about the anchor hole to allow the plurality of bosses to move into the plurality of apertures; and
- a handle extending from the body opposite the anchor arm, such that, when the anchor arm is engaged with the anchor hole the handle provides leverage to increase the available force at the plurality of bosses as the bosses travel into the plurality of apertures.

28. The cleaning tool of claim **27**, wherein the body includes a hinge on a surface opposite the plurality of bosses, and the handle is attached to the body through the hinge such that the handle is moveable between a rest position, wherein the handle extends from the hinge in a direction toward the anchor arm so as to reduce a total length of the cleaning tool, and an operational position, wherein the handle extends from the hinge in a direction opposite the anchor arm, and a storage position.

29. The cleaning tool of claim **27**, wherein each boss in the plurality of bosses is chamfered to allow the boss to engage the respective aperture at an angle while the cleaning tool is being pivoted to move the boss into the aperture.

30. The cleaning tool of claim **27**, wherein the anchor arm includes a hook to allow the anchor arm to engage the anchor hole.

31. In combination, the food processor of claim **19**, and a dicing grid accessory for the food processor to be cleaned by the cleaning tool of claim **27**, the dicing grid accessory having:

a dicing grid with a plurality of apertures; and

an anchor hole located adjacent to the dicing grid.

32. The dicing grid accessory of claim **31**, wherein the anchor hole includes a linear slot.

33. The dicing grid of claim **31**, wherein the anchor hole includes two or more linear slots located along a common anchor direction.

34. The dicing grid accessory of claim **31**, wherein the dicing grid accessory includes an opening for receiving a drive shaft of the food processor, wherein the dicing grid is on a first side of the dicing grid accessory, and wherein the linear slot is on a second side of the dicing grid accessory along an anchor direction.

35. The dicing grid of claim **34**, wherein dicing grid accessory has a circular shape, with the opening in the center, and the anchor direction is a non-diametric chord of the dicing grid accessory.

36. In combination, the food processor of claim **19**, the cleaning tool of claim **27**, and the dicing grid accessory of claim **31**.

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